

SET 3: EMPLOYEE PERFORMANCE EVALUATION

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Course Code & Name	DSA0601 – DATA HANDLING AND VISUALIZATION

DATASET

Employee ID	Department	Years of Service	Performance Score
1	Sales	5	85
2	HR	3	92
3	Marketing	7	78
4	Sales	4	90
5	HR	2	76

QUESTION 1 – LINE CHART: EMPLOYEE PERFORMANCE TREND

AIM

To create a line chart in RStudio showing employee performance scores with labels and a legend.

ALGORITHM

1. Create employee ID and performance score vectors.
2. Use plot() to create a line chart.
3. Add labels and title.
4. Add a legend.
5. Display the graph.

PROCEDURE IN RSTUDIO

Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

R PROGRAMMING CODE FOR RSTUDIO

```
employee_id <- c(1, 2, 3, 4, 5)
performance <- c(85, 92, 78, 90, 76)

plot(employee_id, performance,
      type = "o",
      pch = 16,
      xlab = "Employee ID",
      ylab = "Performance Score",
      main = "Employee Performance Trend")

legend("topright",
      legend = "Performance Score",
      lty = 1,
```

```
pch = 16)
```

```
grid()
```

OUTPUT SCREENSHOT



RESULT

Thus, the employee performance trend was visualized successfully using a line chart in RStudio.

QUESTION 2 – BAR CHART: EMPLOYEES BY DEPARTMENT

AIM

To generate a bar chart in RStudio showing the distribution of employees across departments.

ALGORITHM

1. Create a department vector.
2. Count department values using `table()`.
3. Use `barplot()` to generate the chart.
4. Add labels and title.

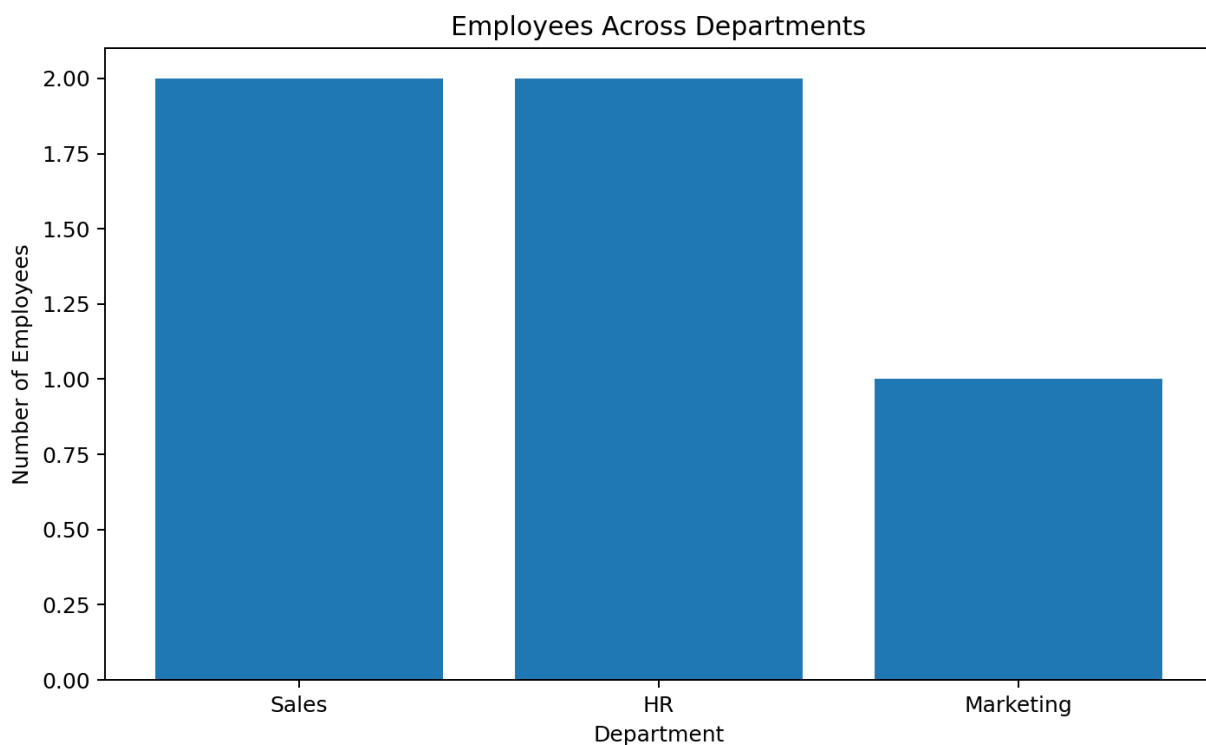
PROCEDURE IN RSTUDIO

Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

R PROGRAMMING CODE FOR RSTUDIO

```
departments <- c("Sales", "HR", "Marketing", "Sales", "HR")  
  
department_counts <- table(departments)  
  
barplot(department_counts,  
        xlab = "Department",  
        ylab = "Number of Employees",  
        main = "Employees Across Departments")
```

OUTPUT SCREENSHOT



RESULT

Thus, the department-wise employee distribution was displayed successfully in RStudio. Sales and HR each contain two employees, while Marketing contains one employee.

QUESTION 3 – SCATTER PLOT: YEARS OF SERVICE VS PERFORMANCE

AIM

To create a scatter plot in RStudio and analyze the correlation between years of service and performance scores.

ALGORITHM

1. Create years of service and performance vectors.
2. Use `plot()` to create the scatter plot.
3. Add labels, title and grid.
4. Calculate the correlation coefficient using `cor()`.
5. Interpret the relationship.

PROCEDURE IN RSTUDIO

Open RStudio. Click File > New File > R Script. Copy the R program into the script editor. Save the script and select the required code. Click Run or press Ctrl+Enter. The graph is displayed in the Plots pane of RStudio.

R PROGRAMMING CODE FOR RSTUDIO

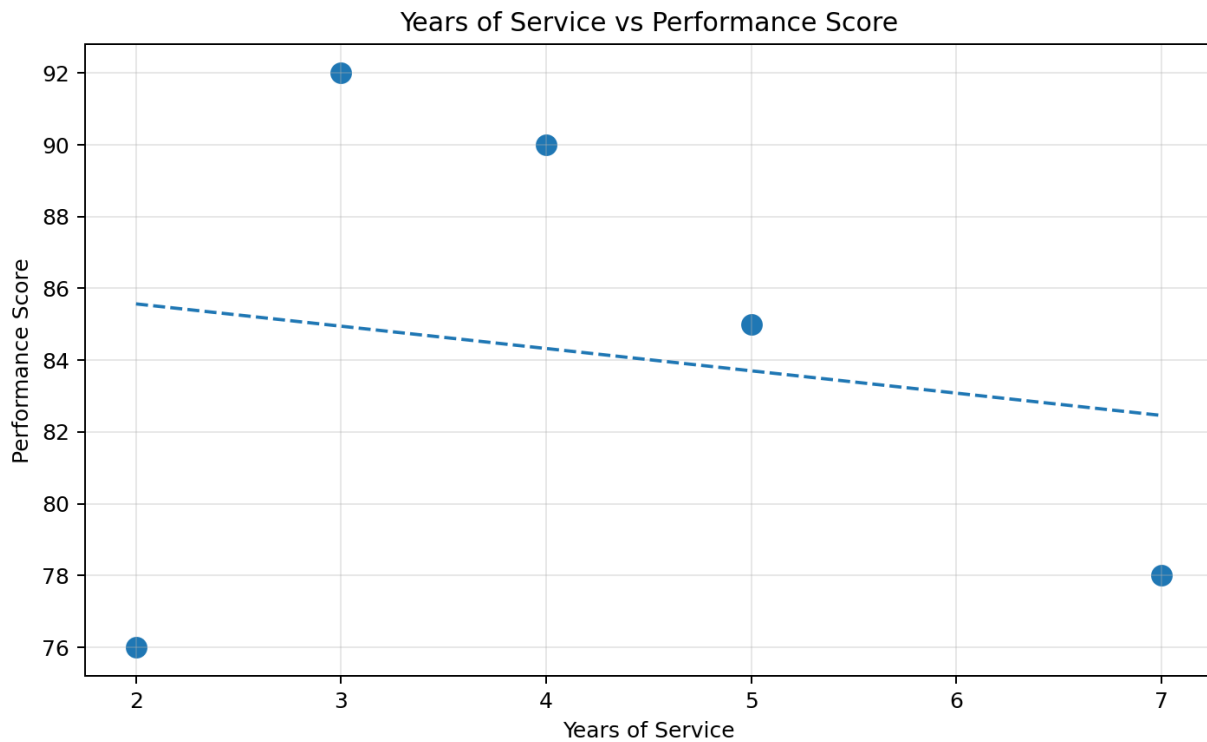
```
years_of_service <- c(5, 3, 7, 4, 2)
performance_score <- c(85, 92, 78, 90, 76)

plot(years_of_service, performance_score,
     pch = 16,
     xlab = "Years of Service",
     ylab = "Performance Score",
     main = "Years of Service vs Performance Score")

grid()

correlation <- cor(years_of_service, performance_score)
print(correlation)
```

OUTPUT SCREENSHOT



RESULT

Thus, the scatter plot was created successfully in RStudio. The sample data does not show a strong positive relationship between years of service and performance.

QUESTION 4 – TABLEAU DASHBOARD: EMPLOYEE PERFORMANCE

AIM

To prepare the employee dataset in RStudio and develop a Tableau dashboard using the line chart and bar chart.

ALGORITHM

1. Create the employee data frame in RStudio.
2. Export the data as a CSV file using write.csv().
3. Connect Tableau to the exported CSV.
4. Create the line chart and bar chart worksheets.
5. Combine both worksheets in a Tableau dashboard.

PROCEDURE IN RSTUDIO

Open RStudio and execute the R code below. The CSV file is saved in the current RStudio working directory. Use getwd() to view the folder. Open Tableau, connect to Employee_Performance.csv, create the two worksheets, and combine them in a new dashboard.

R PROGRAMMING CODE FOR RSTUDIO

```
employee_data <- data.frame(  
  Employee_ID = c(1, 2, 3, 4, 5),  
  Department = c("Sales", "HR", "Marketing", "Sales", "HR"),  
  Years_of_Service = c(5, 3, 7, 4, 2),  
  Performance_Score = c(85, 92, 78, 90, 76)  
)  
  
write.csv(employee_data,  
  "Employee_Performance.csv",  
  row.names = FALSE)  
  
getwd()  
  
# TABLEAU STEPS:  
# Sheet 1:  
# Employee_ID -> Columns  
# Performance_Score -> Rows  
# Marks -> Line  
#  
# Sheet 2:  
# Department -> Columns  
# COUNT(Employee_ID) -> Rows  
# Marks -> Bar  
#  
# New Dashboard -> Add Sheet 1 and Sheet 2
```

OUTPUT SCREENSHOT

Employee Performance Dashboard - Tableau View



Dashboard output preview: line chart and department bar chart combined for interactive exploration

RESULT

Thus, the employee data was prepared in RStudio and exported successfully for developing the interactive Tableau dashboard.